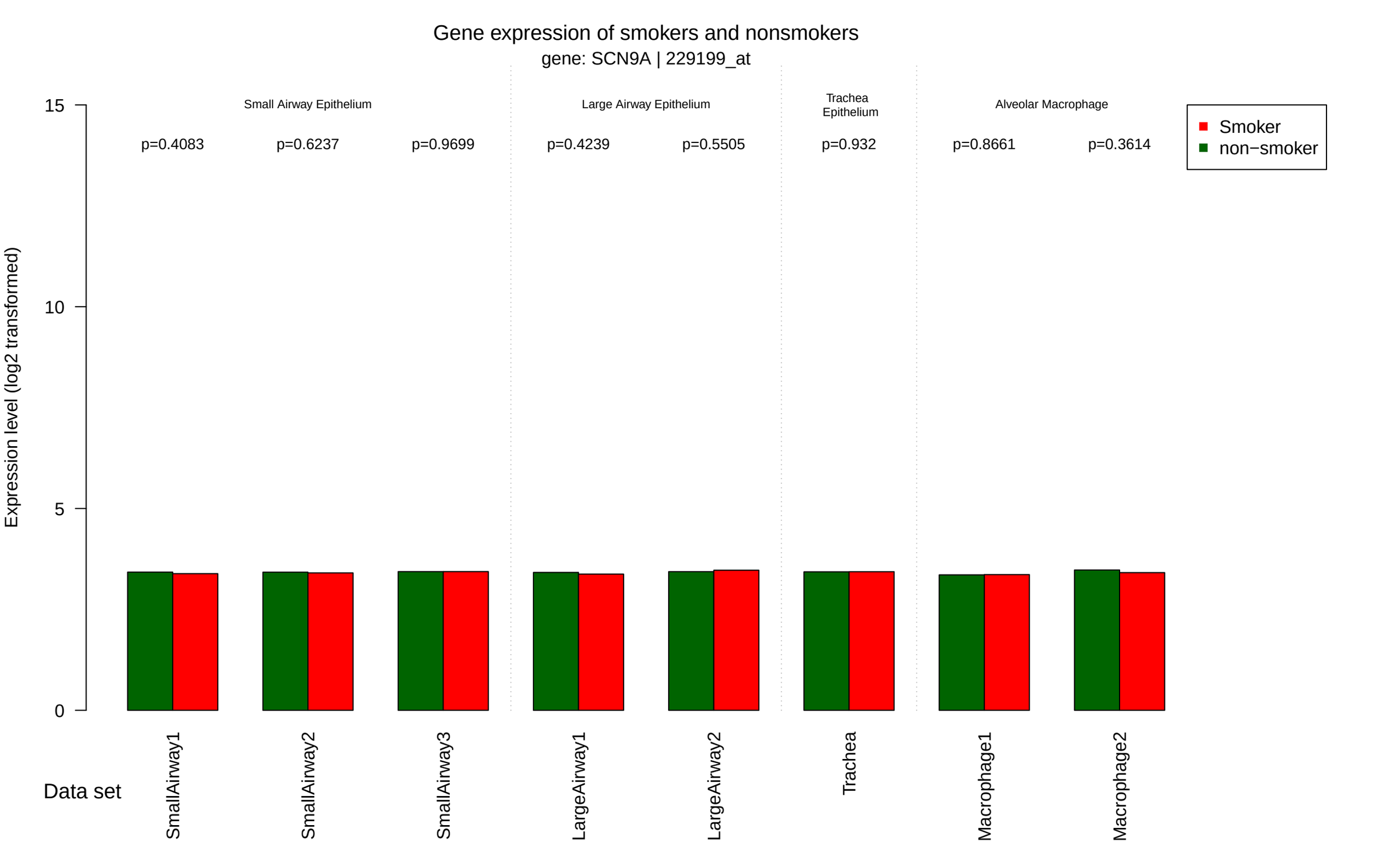
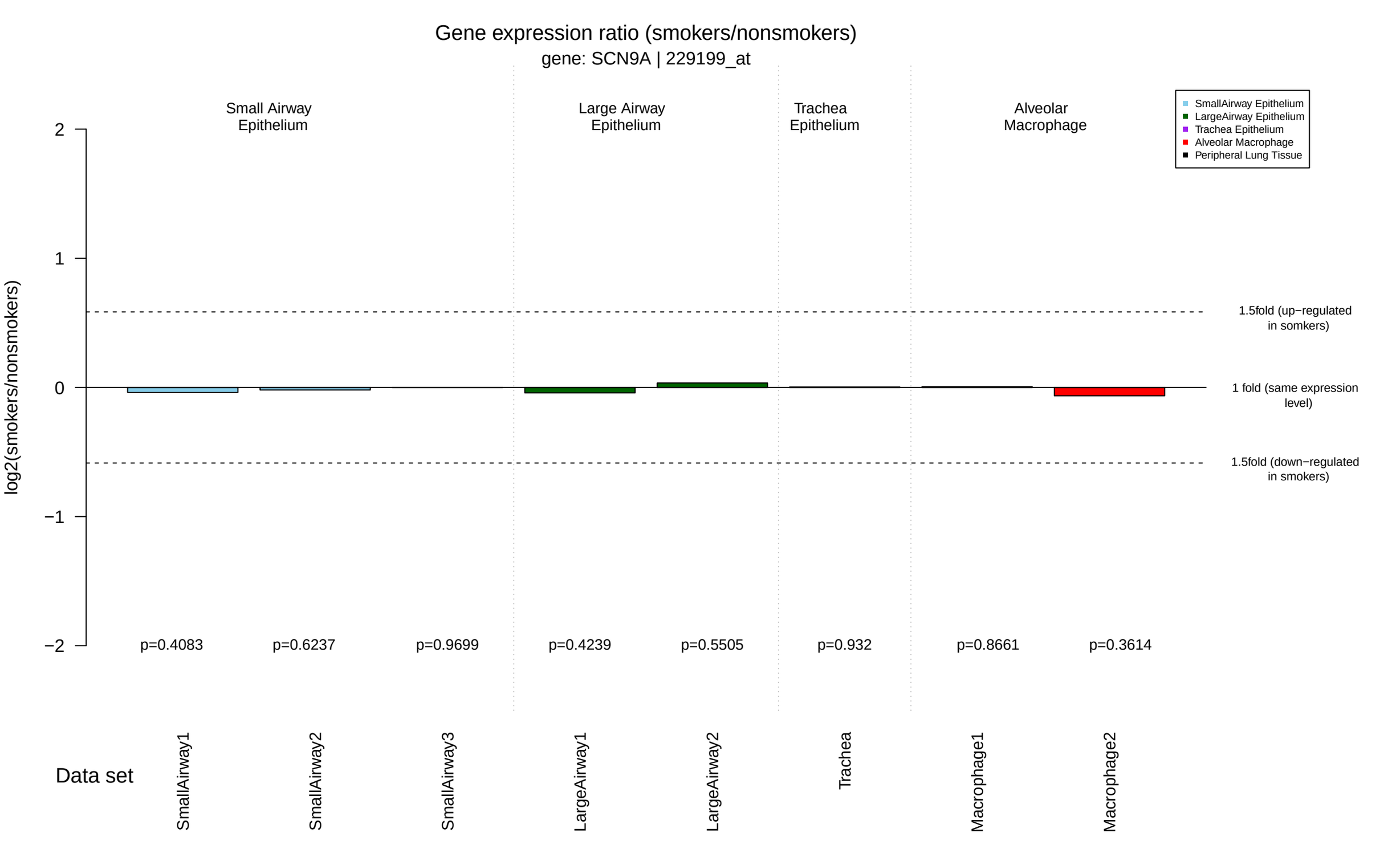
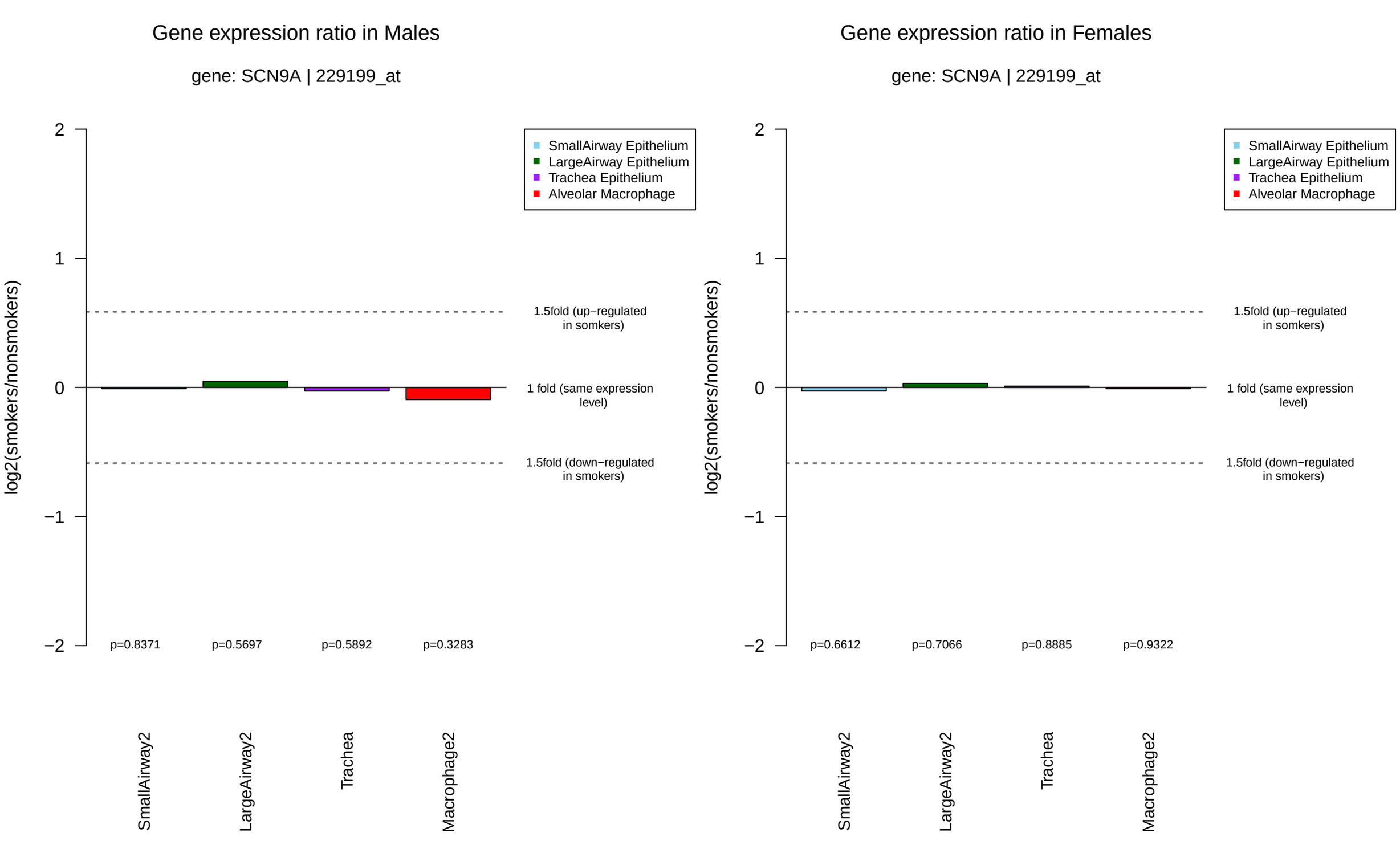
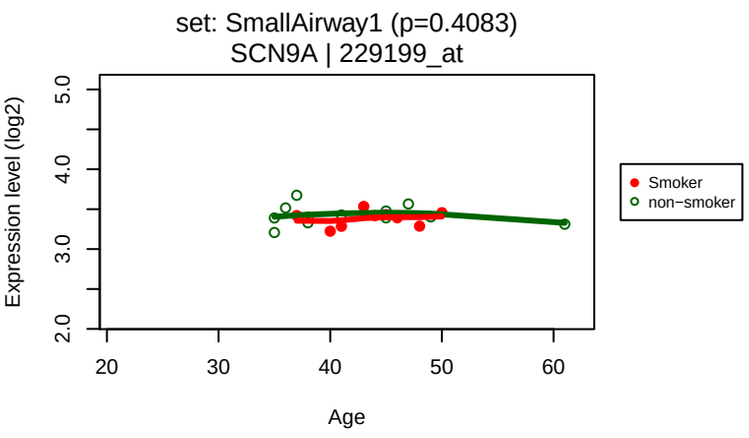
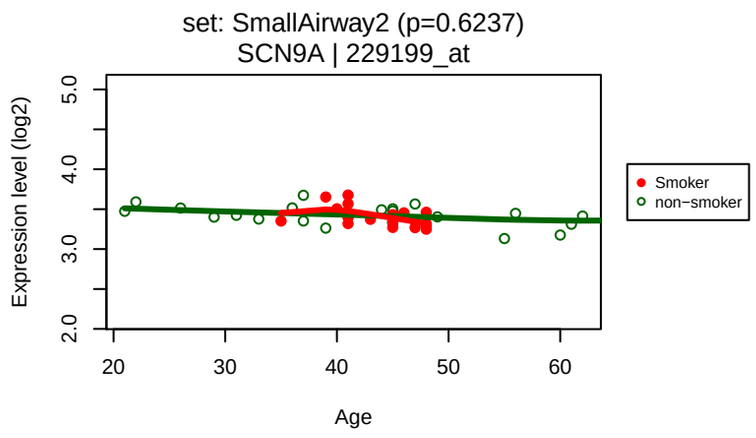
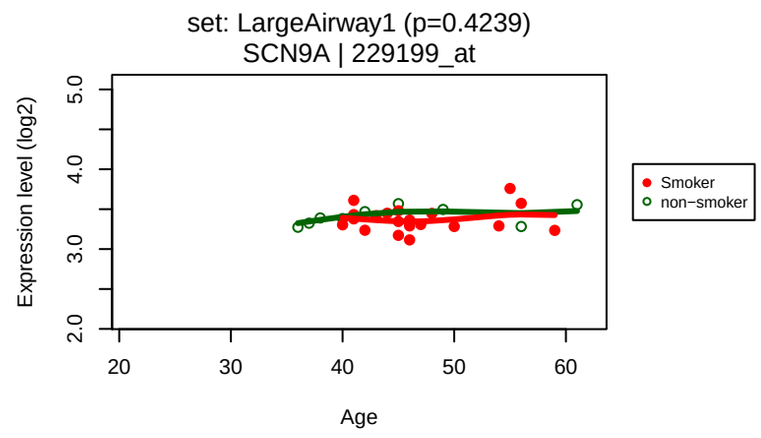

Gene expression of smokers and nonsmokers
gene: SCN9A | 229199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4083
p=0.6237
p=0.9699
p=0.4239
p=0.5505
p=0.932
p=0.8661
p=0.3614
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

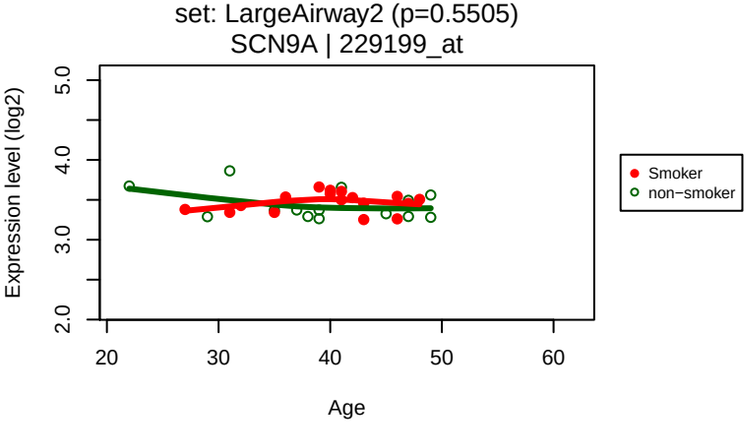

Gene expression ratio (smokers/nonsmokers)
gene: SCN9A | 229199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4083
p=0.6237
p=0.9699
p=0.4239
p=0.5505
p=0.932
p=0.8661
p=0.3614
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

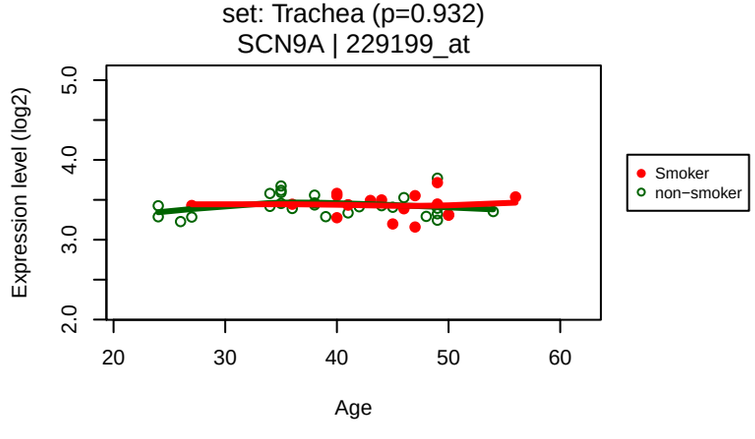

Gene expression ratio in Males
gene: SCN9A | 229199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8371
p=0.5697
p=0.5892
p=0.3283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SCN9A | 229199_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6612
p=0.7066
p=0.8885
p=0.9322
SmallAirway2
LargeAirway2
Trachea
Macrophage2

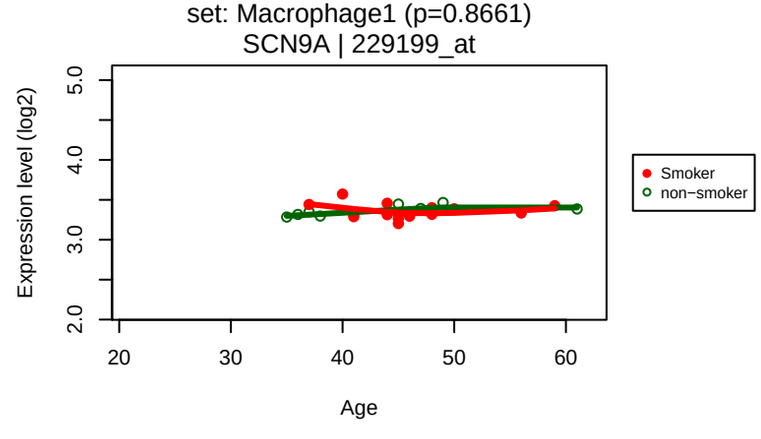

set: SmallAirway1 (p=0.4083)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker

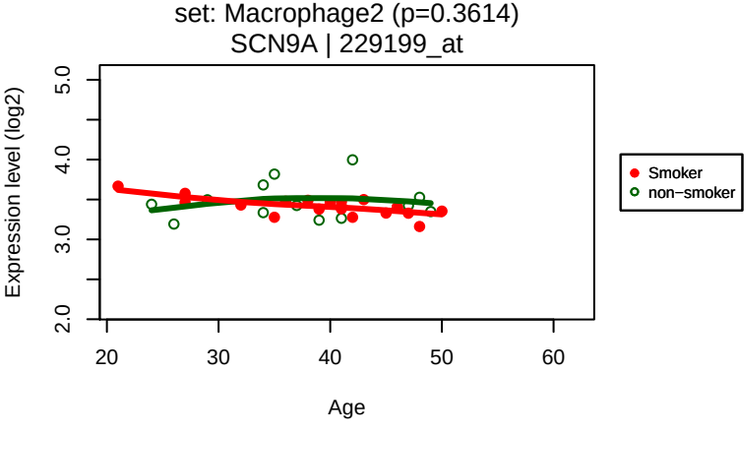

set: SmallAirway2 (p=0.6237)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker

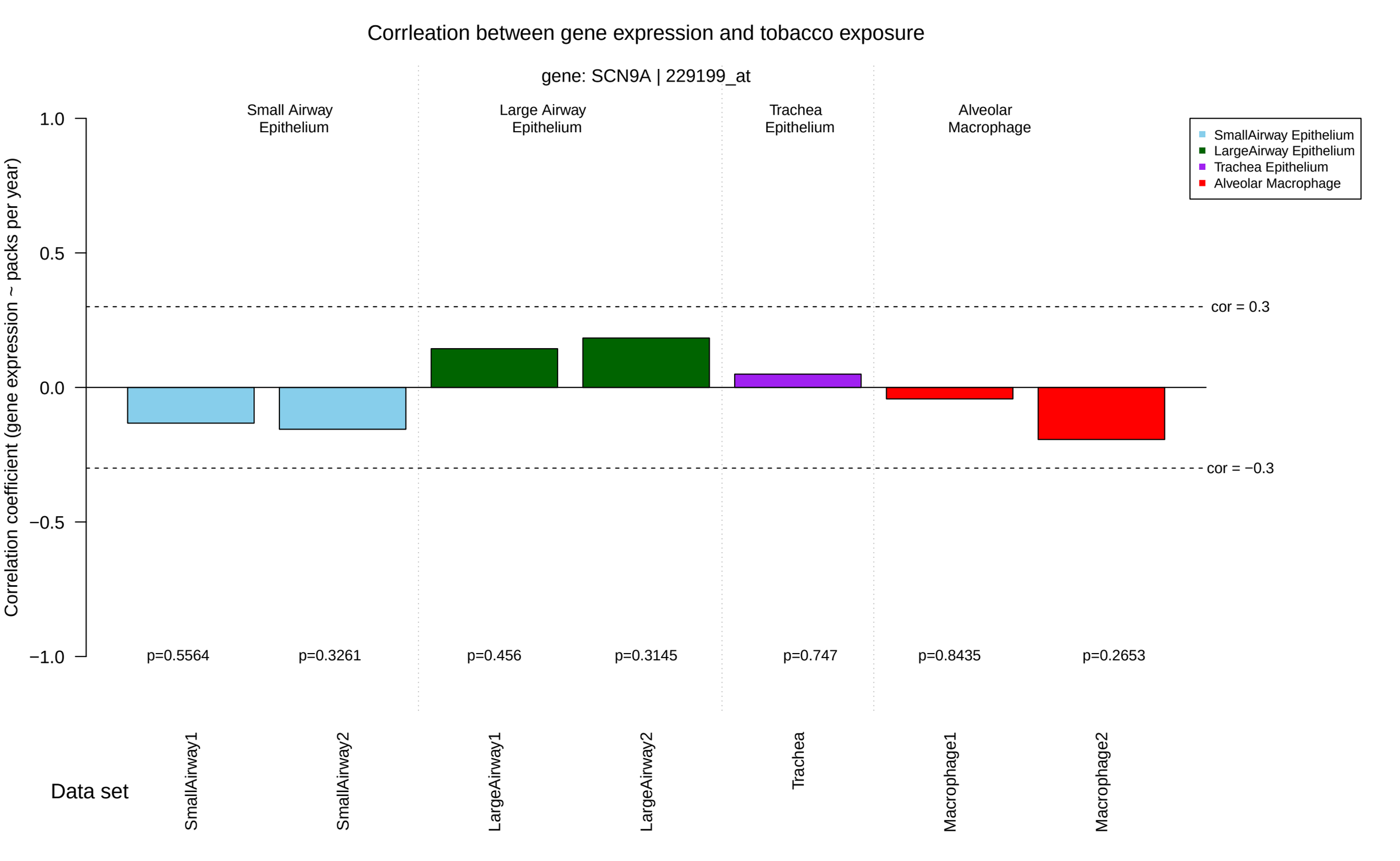

set: LargeAirway1 (p=0.4239)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5505)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.932)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8661)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3614)
SCN9A | 229199_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SCN9A | 229199_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5564
p=0.3261
p=0.456
p=0.3145
p=0.747
p=0.8435
p=0.2653
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2